# CBSE CLASS X • INFORMATION TECHNOLOGY (SUBJECT CODE 402)

## UNIT 1: DIGITAL DOCUMENTATION (ADVANCED) — CHAPTER 2

## WORKING WITH IMAGES — STRUCTURED REVISION & NOTEBOOK NOTES

**Syllabus Sub-Unit 2 Complete Coverage • Concept Bulletins • Practical Steps • Comparison Tables • Zero Over-Shortening**

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### 1. Types of Graphics in LibreOffice Writer & Supported Formats

A Writer document supports two fundamental categories of visual objects to enhance layout clarity and document presentation:

- **Pictures / External Image Files:** Bitmap (raster) or vector illustrations generated outside Writer (via digital cameras, graphic software, scanners, or internet downloads). Stored in standard image formats such as **JPEG/JPG** (Joint Photographic Experts Group), **PNG** (Portable Network Graphics), **GIF** (Graphics Interchange Format), and **BMP**.
- **Drawing Objects:** Native vector shapes drawn directly inside Writer using tools on the Drawing Toolbar (e.g., straight lines, arrows, basic AutoShapes, callouts, text boxes, and Fontwork / Graphic TextArt).
- **Raster vs. Vector Behavior:** Resizing a **Raster image** (composed of fixed pixel grids like JPG/PNG) reduces resolution and causes pixelation or blurring. A **Vector image** (composed of mathematical paths) scales smoothly without loss of sharpness.

### 2. Methods of Inserting Images into a Document

LibreOffice Writer provides five distinct methods to insert an image into a document:

- **Method 1: Insert Image Dialog (Standard Method):** Position cursor → Select **Insert** → **Image...** (or click *Insert Image* icon on Standard Toolbar) → Browse computer folder → Select file → Click **Open**. (By default, the image is embedded and centered above the paragraph).
- **Method 2: Inserting an Image by Linking:** In the *Insert Image* dialog box, locate the image file, check the **Link** checkbox at the bottom-left before clicking **Open**.
- **Method 3: Drag and Drop Method:** Open File Explorer alongside Writer → Click and hold the target image file → Drag it into the document canvas → Release the mouse button at the desired position.
- **Method 4: Linking via Drag and Drop:** Hold down **Ctrl + Shift** simultaneously while dragging and dropping an image file from File Explorer into Writer. This establishes a dynamic file link instead of embedding.
- **Method 5: Copy and Paste (Clipboard):** Select image in File Explorer or image viewer → Copy (**Ctrl + C**) → In Writer document, position cursor → Paste (**Ctrl + V**).
- **Method 6: From Built-in Gallery:** Open Gallery via **Tools** → **Gallery** (or click *Gallery* icon on Sidebar / Standard Toolbar) → Select a theme (e.g., Arrows, Diagrams, Icons) → Drag and drop the graphic into the document.

### 3. Embedding vs. Linking Images (Concept & Comparison)

| Comparison Parameter | Embedding Image (Default) | Linking Image (Linked File) |
|---|---|---|
| **Storage Mechanism** | The complete binary image file is permanently stored and saved **inside** the .odt document file. | Only the **file path reference / pointer** is saved inside the document; the image remains externally on disk. |
| **Document File Size** | Significantly **increases the document size**, especially when multiple high-resolution photos are inserted. | Keeps the document file size **very small and light**, as image data is not duplicated inside the file. |
| **Dynamic Updates** | The document image stays static. Edits to the source file on disk do not reflect in the document. | Any edits or replacements made to the external image file are **automatically loaded** each time the document is opened. |
| **Portability Risk** | Completely self-contained. Moving or emailing the document alone preserves all images without breaking. | If the external image file is moved, renamed, or deleted from its original disk path, the link breaks. |

### 4. Modifying Images: Properties, Resizing, Cropping & Deleting

Selecting an image reveals **eight green sizing handles** and activates the *Image Toolbar* and *Properties Deck* (Sidebar shortcut: **Ctrl + F5** or menu **Format** → **Image** → **Properties...**).

- **Key Visual Properties:**
  - *Brightness:* Regulates the overall lightness or darkness of the image.
  - *Contrast:* Controls the tone difference between light and dark regions.
  - *Color Modes:* **Default** (natural colors), **Grayscale** (256 gray tones), **Black/White** (monochrome high-contrast pixels), and **Watermark** (high brightness, very low contrast, suited for background logos under readable text).
  - *Transparency:* Varies image opacity from 0% (fully opaque) to 100% (fully transparent).
- **Resizing (Scaling) an Image:**
  - *Interactive Dragging:* Click and drag corner sizing handles. **Hold down Shift key** while dragging to maintain the aspect ratio (proportional width-to-height scaling) and prevent distortion.
  - *Exact Precision Sizing:* Right-click image → **Properties...** → **Type tab** (or Properties Deck) → Enter exact Width and Height in cm → Check **Keep ratio**.
- **Deleting an Image:** Click on the image to activate sizing handles → Press the **Delete** key on the keyboard.
- **Cropping an Image:** Cropping means removing unwanted peripheral areas from an image starting from its boundaries without scaling down contents.
  - *Procedure:* Select image → Right-click and choose **Crop** (or click Crop icon on Image Toolbar) → Black crop handles appear → Drag crop handles inward over unwanted margins.

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### Cropping Options: Keep Scale vs. Keep Image Size:

| Feature | Keep Scale (Default Cropping) | Keep Image Size |
|---|---|---|
| **Image Scale / Magnification** | Scale remains 100% unchanged. Cropping simply trims unwanted border areas away. | Remaining visible image is automatically magnified / scaled to fit original frame dimensions. |
| **Physical Frame Size** | The overall physical dimensions (width & height) of the image frame on the page **decrease**. | The overall boundary dimensions of the image frame on the page **remain unchanged**. |

## 5. Graphic Image Filters (Image Toolbar / Format → Image → Filter)

| Filter Name | Action / Visual Effect | Filter Name | Action / Visual Effect |
|---|---|---|---|
| **Invert** | Reverses color values / brightness (photographic negative). | **Aging** | Simulates old photo look with sepia/brown tones. |
| **Sharpen** | **Increases contrast** between adjacent pixels (crisp edges). | **Posterize** | Reduces color depth to make photograph look like a painting. |
| **Smooth** | **Softens contrast** and blends noisy pixels. | **Pop Art** | Converts image into saturated, vibrant pop-art colors. |
| **Remove Noise** | Eliminates isolated single stray pixels (speckles). | **Charcoal Sketch** | Transforms photograph into hand-drawn charcoal sketch. |
| **Solarisation** | Simulates light exposure during chemical film development. | **Relief** | Calculates light source to create 3D embossed shadow look. |
| **Mosaic** | Groups adjacent pixel clusters into rectangular color tiles. | **Flip H / V** | Flips image horizontally (mirror) or vertically (upside down). |

## 6. Working with Drawing Objects & Grouping

- **Activating Drawing Toolbar:** Choose **View** → **Toolbars** → **Drawing** (or click *Show Draw Functions* icon on Standard Toolbar).
- **Creating Drawing Objects:** Click any tool on the Drawing Toolbar (Line, Arrow, Rectangle, Ellipse, Text Box) and drag on the document. Hold **Shift** to draw constrained straight lines, 45° angles, perfect squares, or circles.
- **Setting Object Properties:**
  - *Line Properties:* Select object → Choose **Format** → **Text Box and Shape** → **Line...** (or right-click → Line...). Set line style (solid, dashed), color, width (thickness), and arrowheads.
  - *Area Fill Properties:* Select object → Choose **Format** → **Text Box and Shape** → **Area...** (or right-click → Area...). Select fill style (None, Color, Gradient, Hatching, Bitmap, Pattern) and color.
  - *Rotating Shapes:* Select object → Right-click and choose **Rotate**. Handles turn into **red circular handles**. Drag corner handles to rotate clockwise/counter-clockwise.
- **Grouping Drawing Objects (Concept & Steps):**
  - *Concept:* Grouping combines multiple independent drawing shapes to behave as a single combined object for simultaneous moving, resizing, or formatting.
  - *Step 1:* Click the first drawing shape to select it.
  - *Step 2:* Hold down the **Shift** key and click each additional drawing shape to select all of them.
  - *Step 3:* Choose **Format** → **Group** → **Group** from the Menu bar (or right-click selection → **Group** → **Group**).
  - *To Ungroup Later:* Select the grouped object → Choose **Format** → **Group** → **Ungroup**.

## 7. Positioning Graphics in Text: The Big Four Settings

Precise placement and integration of images with document text is governed by four distinct controls:

| Setting | Definition & Core Function | Available Options & Practical Meaning |
|---|---|---|
| **1. Text Wrapping** | Determines the relation of surrounding document prose to the graphic frame. | • **Wrap Off (None):** Text stops above graphic and resumes below it; sides left blank.<br>• **Page Wrap (Parallel):** Text flows on all four sides of the graphic.<br>• **Optimal Page Wrap (Default):** Intelligent wrapping; skips side wrapping if margin gap < 2 cm.<br>• **Wrap Left (Before):** Text flows only on the left side of the graphic.<br>• **Wrap Right (After):** Text flows only on the right side of the graphic.<br>• **Wrap Through:** Superimposes image over text (text flows behind graphic).<br>• **In Background:** Pushes graphic completely behind text layer. |
| **2. Anchoring** | Acts as the reference datum point for positioning image relative to page elements. | • **To Page:** Graphic remains fixed at an exact page position, unaffected by text flow.<br>• **To Paragraph (Default):** Graphic is bound to a paragraph and moves with it as text shifts.<br>• **To Character:** Graphic is bound to a specific character.<br>• **As Character:** Placed inline like a typed character; directly affects line height (ideal for icons/math symbols). |
| **3. Arrangement** | Controls stacking order (z-index) of overlapping images or drawing shapes. | • **Bring to Front:** Moves object to the absolute top of overlapping stack.<br>• **Send to Back:** Moves object behind all overlapping elements.<br>• **Forward One / Back One:** Moves object up or down by one layer level. |
| **4. Alignment** | Controls horizontal and vertical alignment relative to the anchor. | • **Horizontal:** Left, Centered, Right.<br>• **Vertical:** Top, Center, Bottom. |

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